1 A student is investigating the speed of a vehicle on a track when a small ball is projected into the vehicle, as shown in Fig. 1.1.

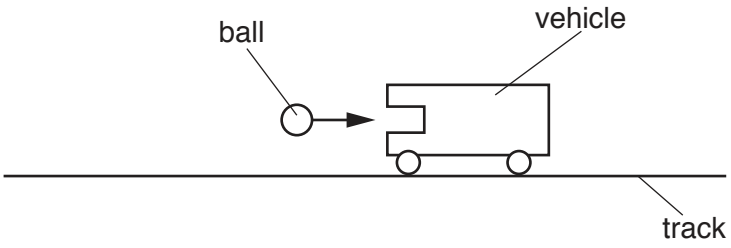


Fig. 1.1

The ball is projected towards the vehicle by a compressed spring. It is suggested that the relationship between the speed v of the vehicle and its mass M, after the ball embeds itself in the vehicle, is

$$kx^2 = (M + b)v^2$$

where b is the mass of the ball, k is the spring constant and x is the compression of the spring.

Design a laboratory experiment to test the relationship between v and M.
Explain how your results could be used to plot a graph with $1/v^2$ on the y-axis and to determine a value for k. You should draw a diagram, on page 3, showing the arrangement of your equipment.

In your account you should pay particular attention to

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

[15]

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